SEONGHEON PARK

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RESEARCH INTERESTS

Foundation Models (LLMs, VLMs, and VLAs); Agentic AI; Robot Manipulation and Embodied AI; Reliable and Safe AI (Failure Detection and Mitigation, Uncertainty Quantification, and Learning from Failure)

EDUCATION

University of Wisconsin-Madison *Sep. 2024 - Present*
Ph.D. in Computer Science
Advisor: Prof. Sharon Li

Yonsei University *Sep. 2021 - Aug. 2023*
M.S. in Electrical and Electronic Engineering
Advisor: Prof. Kwanghoon Sohn

Yonsei University *Mar. 2015 - Aug. 2021*
B.S. in Electrical and Electronic Engineering

PROFESSIONAL EXPERIENCE

Google, New York City, United States *Sep. 2026 – Present*
Student Researcher (Self-Improving Agents)

Microsoft Research Asia, Tokyo, Japan *May 2026 – Aug. 2026*
Research Intern (Robotics, VLA)

WORKING PAPERS

19. Shulin Tian, Shoukang Hu, Heecheol Kim, **Seongheon Park**, Yukang Cao, Namiko Saito, Yasuyuki Matsushita, and Ziwei Liu, "RoboMemo: Semantic-Guided Online Memory for Long-Horizon Vision-Language-Action Policies," Under Review (ICRA)

18. **Seongheon Park**, Heecheol Kim, Shulin Tian, Lilika Makabe, Namiko Saito, Katsushi Ikeuchi, Sharon Li, and Yasuyuki Matsushita, "DiVeR: Decision-Critical Verifier Learning for VLA Test-Time Scaling," **NeurIPS 2026 Workshop** on PTA: From Pretrained Representations to Acting Agents, Under Review (ICLR)

17. Chaeyun Kim, Jongwook Han, Jeongeun Park, Kihyun Kim, **Seongheon Park**, and Haon Park, "Risks Beyond Actions: Contextual Trajectory Safety for Embodied AI," **ECCV 2026 Workshop** on Safe World Models for Trustworthy Embodied AI

PUBLICATIONS

16. Changdae Oh, Wendi Li, **Seongheon Park**, Samuel Yeh, Tanwi Mallick, and Sharon Li, "Neglected Free Lunch from Post-training: Progress Advantage for LLM Agents," In **NeurIPS 2026**

15. **Seongheon Park**, Wendi Li, Changdae Oh, Samuel Yeh, Zsolt Kira, Michael Hagenow, and Sharon Li, "Hide-and-Seek in Trajectories: Discovering Failure Signals for VLA Runtime Monitoring," In **NeurIPS 2026**

14. Changdae Oh, **Seongheon Park**, To Eun Kim, Jiatong Li, Wendi Li, Samuel Yeh, Xuefeng Du, Hamed Hassani, Paul Bogdan, Dawn Song, and Sharon Li, "Uncertainty Quantification in LLM Agents: Foundations, Emerging Challenges, and Opportunities," In **ACL 2026**

13. **Seongheon Park**, Changdae Oh, Hyeong Kyu Choi, Xuefeng Du, and Sharon Li, "VAUQ: Vision-Aware Uncertainty Quantification for LVLM Self-Evaluation," In **ACL 2026**

12. Lin Long*, Changdae Oh*, **Seongheon Park**, and Sharon Li, "Understanding Language Prior of LVLMs by Contrasting Chain-of-Embedding," In **ICLR 2026**

11. **Seongheon Park** and Sharon Li, "GLSim: Detecting Object Hallucinations in LVLMs via Global-Local Embedding Similarity," In **NeurIPS 2025**

10. Pengyue Jia, **Seongheon Park**, Song Gao, Xiangyu Zhao, and Sharon Li, "GeoRanker: Distance-Aware Ranking for Worldwide Image Geolocalization," In **NeurIPS 2025**

9. Jinyuan Luo, Zhen Fang, Sharon Li, **Seongheon Park**, and Ling Chen, "Shaking to Reveal: Perturbation-Based Detection of LLM Hallucinations," arXiv:2506.02696

8. Min-Hsuan Yeh, Jeffrey Wang, Xuefeng Du, **Seongheon Park**, Leitian Tao, Shawn Im, and Sharon Li, "Position: Challenges and Future Directions of Data-Centric AI Alignment," In **ICML 2025**

7. **Seongheon Park**, Xuefeng Du, Min-Hsuan Yeh, Haobo Wang, and Sharon Li, "Steer LLM Latents for Hallucination Detection," In **ICML 2025**

6. Min-Hsuan Yeh, Max Kamachee, **Seongheon Park**, and Sharon Li, "HalluEntity: Benchmarking and Understanding Entity-Level Hallucination Detection," In **TMLR, ICLR 2026**

5. **Seongheon Park***, Hyuk Kwon*, Kwanghoon Sohn, and Kibok Lee, "Rethinking Open-World Semi-Supervised Learning: Distribution Mismatch and Inductive Inference," In **CVPR 2024 Workshop**

4. Hanjae Kim, Jiyoung Lee, **Seongheon Park**, and Kwanghoon Sohn, "Hierarchical Visual Primitive Experts for Compositional Zero-Shot Learning," In **ICCV 2023**

3. Minsu Kim, Seungryong Kim, Jungin Park, **Seongheon Park**, and Kwanghoon Sohn, "PartMix: Regularization Strategy to Learn Part Discovery for Visible-Infrared Person Re-identification," In **CVPR 2023**

2. **Seongheon Park**, Hanjae Kim, Minsu Kim, and Kwanghoon Sohn, "Normality Guided Multiple Instance Learning for Weakly Supervised Video Anomaly Detection," In **WACV 2023 (Oral)**

1. Dahye Kim, Jiyoung Lee, Jungin Park, **Seongheon Park**, and Kwanghoon Sohn, "Language-free Training for Zero-shot Video Grounding," In **WACV 2023 (Oral)**

RESEARCH EXPERIENCE

University of Wisconsin–Madison, USA *Sep. 2024 - Present*
Research Assistant (Advisor: Prof. Sharon Li)

Machine Learning Lab, Yonsei University, South Korea *Sep. 2023 - Aug. 2024*
Research Assistant (Advisor: Prof. Kibok Lee)

Digital Image Media Lab, Yonsei University, South Korea *Sep. 2021 - Aug. 2023*
Research Assistant (Advisor: Prof. Kwanghoon Sohn)

Digital Image Media Lab, Yonsei University, South Korea *Jan. 2021 - Aug. 2021*
Undergraduate Research Intern (Advisor: Prof. Kwanghoon Sohn)

Multi-Dimensional Insight Lab, Yonsei University, South Korea *Jan. 2019 - Mar. 2019*
Undergraduate Research Intern (Advisor: Prof. Sanghoon Lee)

MILITARY SERVICE

Republic of Korea Army Logistics Command *Jul. 2019 - Jan. 2021*

TEACHING EXPERIENCES

- EEE 3313: Introduction to Digital Labs *Fall 2022*
- EEE 4420: Digital Signal Processing *Spring 2022*
- EEE 6321: Image Coding *Spring 2022*

PROJECTS

1. **Project with Korea Institute of Science and Technology (KIST)** *Jan. 2022 - Aug. 2023*
   - Project Title : Deep Identification and Tracking of Missing Persons in Heterogeneous CCTV
   - Covers : Object Detection/Tracking, Video Action Classification, Person Re-Identification, Anomaly Detection
2. **Project with Ministry of Science and ICT, Mid-Level Research** *Jan. 2022 - Aug. 2023*
   - Project Title : Development of Complex Situational Awareness and Prediction Technology through Multi-Modal Data Fusion and Social Artificial Intelligence
   - Covers : Multi-Modal Learning, Video Understanding
3. **Project with Yonsei University-Yonsei Signature Research Cluster** *Jan. 2022 - Jan. 2023*
   - Project Title : Development of Multimodal-based General-purpose Social Artificial Intelligence Technology
   - Covers : Multi-Modal Learning, Zero-Shot Learning, Meta-Learning

PATENTS

1. Korean Patent, No.10-2022-0156145 (Anomaly Detection)
2. Korean Software, No.2022-0049 (Anomaly Detection)

HONORS AND SCHOLARSHIPS

Honors

- Honors Student, Yonsei University *2016, 2018-2019*

Scholarships

- Japanese Government (MEXT) Scholarship for Doctoral Studies,
  The University of Tokyo (selected; declined) *2024*
- Teaching Assistant Scholarship, Yonsei University *2022*
- Research Assistant Scholarship, Yonsei University *2021 - 2023*
- Brain Korea 21 (BK21) Scholarship , National Research Foundation of Korea *2021 - 2023*
- Future Vehicula Technology Scholarship, Inter-University Alliance *2021 - 2023*
- Academic Excellence Scholarship, Yonsei University *2019*

LANGUAGE

Korean (Native), English (Fluent), Japanese (Advanced, JLPT N1)